Midterm Project Report

Supervised Learning - 2024-2025

Project Topic: Analysis and Prediction of Weight Loss in Subjects Undergoing Diet

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The dataset used in this project consists of 14 features (9 numerical and 5 categorical) with a total of 600 rows (samples). The target variable is "Final Weight (lbs)".

1. Processing data

a) Symbol '-': In the dataset, the symbol '-' was present in the **Smoke** and **Gender** features. For **Smoke**, the symbol was replaced with **'NO'** since only the value **'yes'** was present, indicating that the symbol likely meant **'NO'**. For **Gender**, the symbol was replaced with **NaN** to avoid introducing biases. The analysis suggested that the dataset was divided into male and female, but to be cautious, these entries were marked as **NaN**.

b) Outliers/Missing Values: In the dataset, a small number of **outliers** were identified. After analysis, these were replaced with **NaN**. Specifically, **10 outliers** in the **'Age'** feature were identified as noise (values around 200 or negative). The remaining **5 outliers** were also set to NaN to avoid negatively impacting the model's performance.

c) Correlation analysis: For this analysis, a **heatmap** was first plotted to visualize the correlation between the features, revealing **2 significant correlations**. To further analyze the correlation between the features and the target variable, a series of **scatter plots** were used. These plots showed that **BMR (Calories)** has a very strong correlation with the target, indicating its importance for training our models.

d) Features Distribution: The PDF analysis of the features revealed that most **numerical features** followed a distribution similar to a Gaussian, except for **"Daily Calories Consumed"**, which showed a **left-skewed distribution**. To make this feature more Gaussian-like, the following transformations were applied: first, an inversion 1/(x+1) was applied, followed by a log transformation. This did not produce a normal distribution, bat was retained trasformed because improve the performance of the final models.For the **categorical features**, a potential issue was identified with **"Sleep Quality"**, where the **"Excellent"** category had only about 20 entries. This small sample size could lead to generalization issues for the model. To test this, "Excellent" was combined with "Very Good," but this resulted in worse model performance, so the original categories were retained.

e) Scaler features: To improve the dataset, **scaling** was applied to all numerical features. Specifically, a **Min-Max scaler** was used for the **'Daily Calories Consumed'** feature since it does not follow a normal distribution and is very good for knn imputation, while a **Standardization scaler** was applied to the rest of the features due to their symmetric distribution.

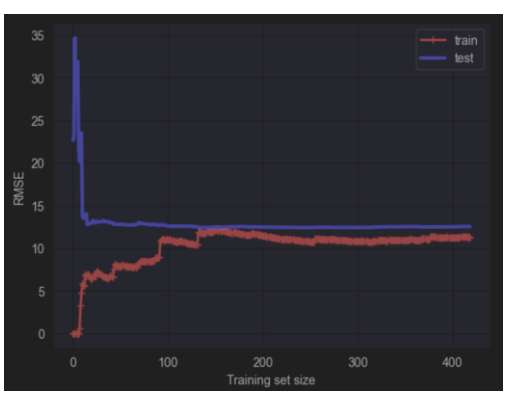
f) Encoding features: A total of **three types of encoding** were used: **Binary encoding** for the features **'Gender'** and **'Smoking'**; **Ordinal encoding** for **'Physical Activity Level'** and **'Sleep Quality'** since they follow a specific order; and **One-Hot Encoding** for the **'Work Sector'** feature, which does not have any inherent order.

g) Features Selection: For feature selection, **three methods** were used: **Univariate (SelectPercentile)** with a **50% threshold** to reduce model complexity, **Iterative (RandomForest)** selecting the top **7 features**, and **Model-Based (Linear Regression)**. To combine these methods, only the features present in at least **two out of the three** were kept. This resulted in a final set of **7 features**.
{'Age_scaled', 'BMR_scaled', 'Daily_Caloric_S/D_scaled', 'Daily_Calories_Consumed_scaled', 'Duration_scaled', 'Gender','Weight_Change_scaled'}

2. Analisys Results:

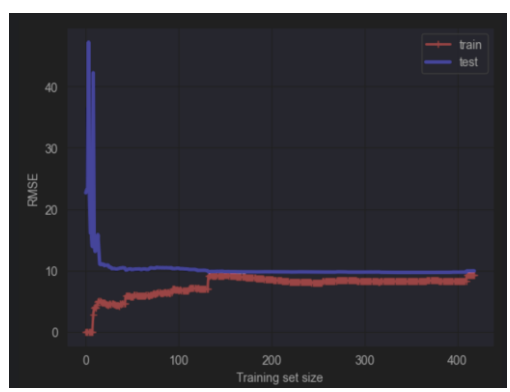
The project involves processing a dataset to optimize it and then developing a series of models to predict the **Final Weight (lbs)** feature. A total of 5 models were developed: **Linear-Regression** using Sklearn, **Batch Gradient Descent**, **Mini-Batch Gradient Descent**,Linear Model with **Lasso Regularisation** and **Ridge Regression Model.** Additionally, some of these models were tested on different datasets. The main dataset was first split into two parts: one including the **BMR** feature and one excluding it. It was further divided based on **gender** (male and female).

(a) Linear Regression with Reduced_dataset:



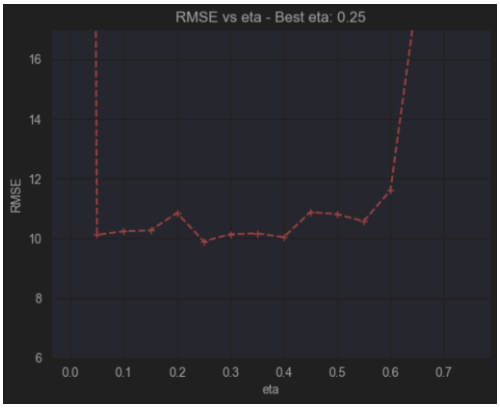
Dataset	Metric	Train	Test
With BMR	RMSE	9.180	9.930
With BMR	R^2	0.776	0.796
Without BMR	RMSE	11.270	12.520
Without BMR	R^2	0.664	0.677

With BMR

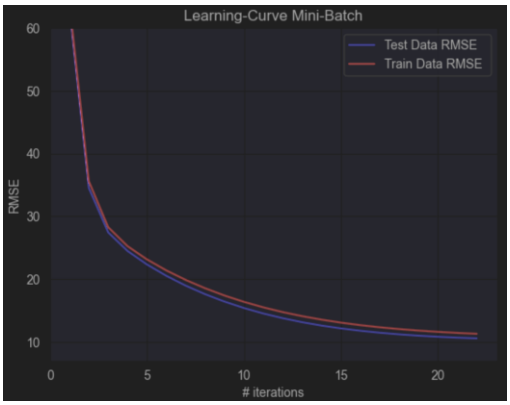


Analyzing these values, we can observe that the model using the **BMR** feature has a **lower RMSE** and a **higher R^2**, indicating better performance and generalization. This improvement can be attributed to the fact that the **BMR** feature has a strong correlation with the target variable, as identified during the dataset processing. Analyzing the two learning curves, we can observe that both models **generalize well** and are not subject to **underfitting** or **overfitting**. This is evident as the **test error decreases** while the **train error increases** as the training set size grows.

(b) Mini Batch Gradient Descent Reduced_dataset: With BMR

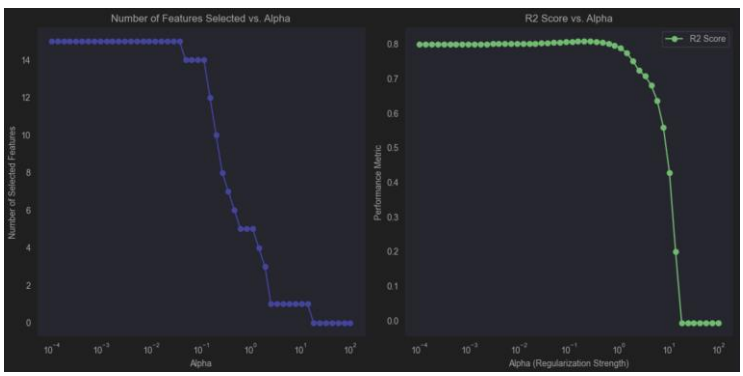


Dataset	Metric	Test Set	Training Set
With BMR	Optimal eta	0.25	-
	Early Stopping	2	-
	RMSE	9.90	9.63
	R²	0.79	0.75
Without BMR	Optimal eta	0.25	-
	Early Stopping	35	-
	RMSE	12.70	11.79
	R²	0.66	0.63



We can observe how the optimal **'eta'** (learning rate) is found during training. As seen in the graph, the **RMSE** value fluctuates up and down, which is typical of the **Mini-Batch Gradient Descent** method.
Regarding the **learning curves**, we can conclude that the model is functioning correctly without signs of **overfitting** or **underfitting**:
Underfitting occurs when both the training and test errors start increasing together.
Overfitting is indicated when the **test error decreases** while the **training error increases**.
As for the performance metrics, we can see that excluding the **BMR** feature results in a slightly worse model, as evidenced by the increased **RMSE** and decreased $\mathbf{R^2}$ values.Another observation is that the **training curve** is above the **test curve**, which could be due to the **random split** of the data into **X_train** and **X_test**. This can occasionally lead to a training set that is slightly easier to predict than the test set, depending on how the split is performed.
For the dataset **without BMR**, the graphs are similar but have **different values**.

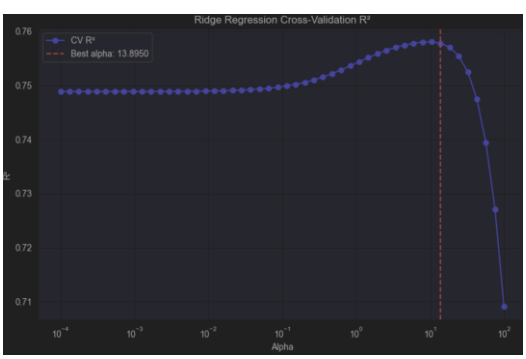
(c) Linear Reg. with Lasso Regularization aumented_dataset:



Dataset	Best Alpha	Best RMSE	Best R²
With BMR	0.2	9.64	0.80
Without BMR	0.0001	12.36	0.685

The analysis conducted using Lasso regularization on datasets, with and without BMR, reveals significant differences in performance.
With BMR, the model achieves an optimal alpha of 0.2, reducing the **RMSE** to 9.64 and reaching an $\mathbf{R^2}$ of 0.80. This indicates that BMR effectively enables the model to eliminate irrelevant features, maintaining high accuracy while reducing complexity.
In contrast, **without BMR**, the model requires a very low alpha (0.0001) to achieve the best performance, resulting in a higher **RMSE** (12.36) and a lower $\mathbf{R^2}$ (0.685). This suggests that, without BMR, the dataset contains many irrelevant or redundant features, making the model less robust to regularization.
In summary, the use of BMR allows for better regularization, reducing the number of features without compromising accuracy, whereas without BMR, the model becomes more complex and less effective.

(d) Ridge Reg. Model aumented_dataset:

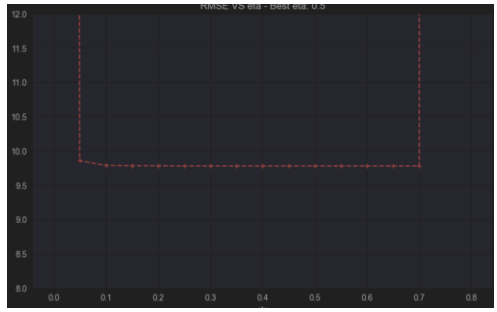


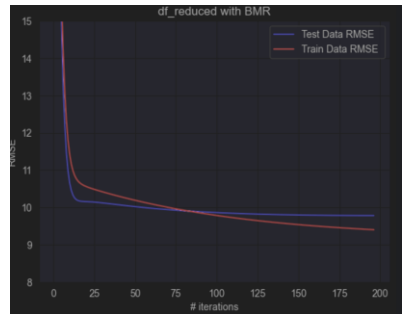
Dataset	Best Alpha	Training RMSE	Training R²	Test RMSE	Test R²
With BMR	13.8950	9.1296	0.7792	9.8004	0.8020
Without BMR	0.0494	11.0900	0.6742	12.3685	0.6846

A plot was created for the $\mathbf{R^2}$ of the Ridge model to show how well the model explains the variability in the data.For **low alpha values**, the $\mathbf{R^2}$ remains constant around **0.75**. As alpha approaches **1**, the R^2 starts to increase, reaching its optimal value before dropping sharply.
Explanation of the Trend
- For **low alpha values**, there is minimal regularization, so the model behaves like standard linear regression.
- As **alpha increases**, moderate regularization helps reduce overfitting, improving R^2 by allowing better generalization.
- Beyond a certain point, **high alpha values** over-penalize the coefficients, leading to **underfitting**. This is why the R^2 drops sharply.

(e) Batch Gradient Model reduced_dataset:

Eta	RMSE Train	RMSE Test	R2 Test	R2 Train	Dataset
0.5	9.39	9.77	0.8	0.76	With BMR
0.65	11.27	12.52	0.67	0.66	Without BMR

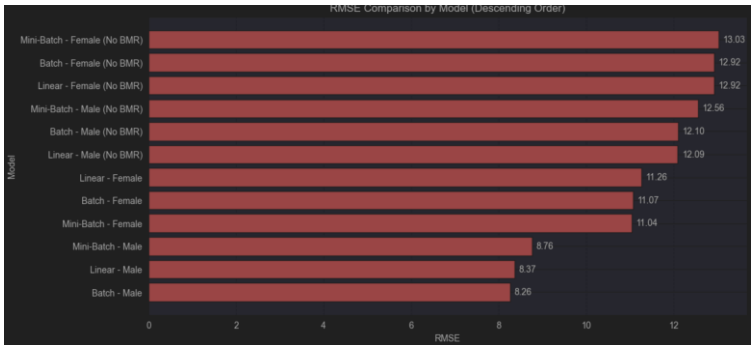




The learning curve shows the **RMSE** for both the training (red) and test (blue) sets over iterations. 1. **Rapid initial convergence**: Both curves decrease quickly in the first 20-30 iterations, indicating the model is learning effectively. 2. **Stable convergence** after ~50 iterations: The curves flatten out. 3. **Small gap** between training and test curves: This indicates good generalization. 4. **Asymptote around RMSE 9-10**: The model reaches its performance limit, with little improvement beyond this point.

(f) Final Results:

The final analysis involves comparing the different models applied to the **reduced dataset**, which are trained on various subsets (**Female without BMR, Female with BMR, Male with BMR, Male without BMR**) to understand how these modifications affect model performance. To achieve this, several **plots and tables** were generated:

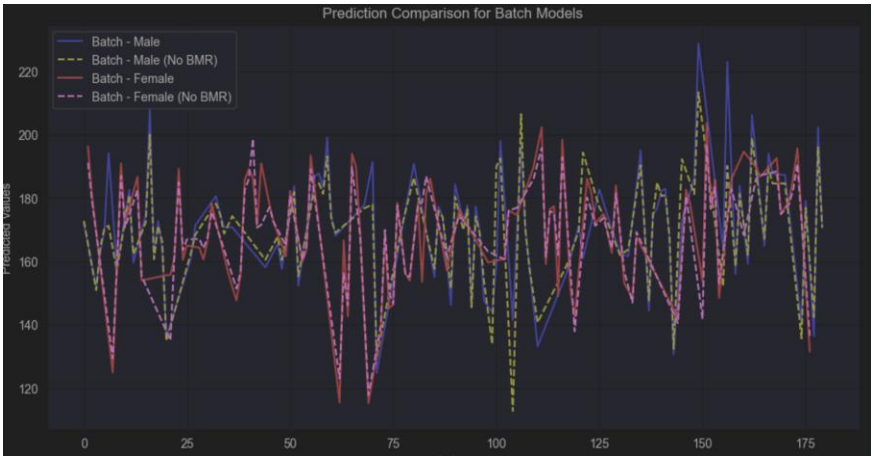


From this **bar plot**, we can identify which specific models perform better by analyzing the **RMSE**. As observed, the top performers are **Batch-Male** and **Linear -Male**. Another key observation is that all models **without the BMR feature** perform worse compared to those with it. Additionally, the the models for Female perform worse compared to Male. This is probably because the dataset of male have more general aspect and complet information.

Mean RMSE for Models With and Without BMR:

	Type	Group	RMSE
0	Batch	No BMR	12.509913
1	Batch	With BMR	9.662491
2	Linear	No BMR	12.507206
3	Linear	With BMR	9.812566
4	Mini-Batch	No BMR	12.791445
5	Mini-Batch	With BMR	9.900732

The key aspect to analyze here is the **average performance** of the different models by combining the **RMSE for Female and Male**. The results indicate that, for the **Batch and Mini-Batch models and Linear**, training separate models for males and females, applying these models to **y_f** and **y_m**, and then merging the results can lead to improved model performance (points **(a), (b), (e)**).



This plot shows the different **y_pred** values, allowing us to make some observations. The plot was created for the **three different models**, which display a very similar pattern across the **four lines**. This indicates that the models predict the various samples in a similar manner. However, what changes is the type of **dataset used for training**. As observed in the plot, predictions vary depending on whether the **BMR feature** is included and based on **gender**, which impacts the model's predictions.